Stewart Bahrke

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Department of Mathematics
University of North Texas
Denton, Texas, USA

RESEARCH INTERESTS

Mathematical logic, set theory, combinatorial set theory, descriptive inner model theory. Currently I am researching generalizations of classical combinatorial constructions, particularly independent families, under the axiom of determinacy, its strengthenings, and sometimes in inner models of determinacy.

EMPLOYMENT

Doktorand
Technische Universität Wien (TU Wien), 2026 – 2027
Teaching Fellow
University of North Texas, 2024 – 2026
Teaching Assistant
University of North Texas, 2022 – 2023
Teaching Assistant
University of Michigan, 2019 – 2022

EDUCATION

Ph.D. in Mathematics
University of North Texas, Fall 2022 – Current
B.Sc. in Mathematics (with Honors)
University of Michigan, 2022

GRADUATE ADVISORS & SPONSORS

Nam Trang, University of North Texas; Denton, Texas, USA (Thesis Advisor)
Sandra Müller, Technical University of Vienna; Vienna, Austria (Marshall Plan Sponsor)

GRANTS, AWARDS, & FUNDING

Simons Semester at IMPAN funding through the Simon's Foundation (June 2026)
Austrian Marshall Plan, Project title: Combinatorial Families Above Θ (2025 – 2026)
R. Mauldin Scholar (2022 – 2023)
Bastings Scholar (2018 – 2022)

TEACHING

Instructor of Record (University of North Texas)

- Math 1710: Calculus I; Fall 2024, Spring 2025, Fall 2025
- Math 2730: Multivariable Calculus; Summer 2024, Summer 2025
- Math 2700: Linear Algebra; Spring 2024

Teaching Assistant (University of North Texas)

- Math 1710: Calculus I; Spring 2023, Fall 2023, Spring 2026
- Math 1100: Algebra; Fall 2022

Teaching Assistant (University of Michigan)

- EECS 203: Discrete Mathematics; Fall 2019, Winter 2020, Fall 2020, Winter 2021, Summer 2021, Fall 2021, Winter 2022

INVITED TALKS	Independent Families on $\kappa < \Theta$, June 2025, Berkeley Inner Model Theory Conference, Berkeley, California. Generalized Cardinal Characteristics and Determinacy, April 2026, UNT Logic Seminar, Denton, Texas Independent Families and Determinacy, November 2026, University of Vienna Set Theory Seminar, Vienna, Austria
PUBLICATIONS AND PREPRINTS	Bahrke, S., Trang, N., *Independent families under determinacy–like hypotheses*, 2026, Submitted to the Proceedings of the American Mathematical Society.
SERVICE	Organizer of University of North Texas Graduate Logic Seminar, Spring 2024, Fall 2025